

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical
Statistics

May 2010 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1) (a) Arranging the data in ascending order

199, 201, 203, 206, 209, 210, 210, 212, 215, 215, 216, 218, 218, 230, 240

Here, $n = 15$.

One can use,

$$Q_1 = \left(\frac{n+2}{4}\right)^{th} \text{ value} = 4\frac{1}{4}^{th} \text{ value} = 206.75$$

$$Q_3 = \left(\frac{3n+2}{4}\right)^{th} \text{ value} = 11\frac{3}{4}^{th} \text{ value} = 217.50$$

$$\text{Inter Quartile Range} = Q_3 - Q_1 = 10.75.$$

Other can use,

$$Q_1 = \left(\frac{n+1}{4}\right)^{th} \text{ value} = 4^{th} \text{ value} = 206.00$$

$$Q_3 = \left(\frac{3n+3}{4}\right)^{th} \text{ value} = 12^{th} \text{ value} = 218.00$$

$$\text{Inter Quartile Range} = Q_3 - Q_1 = 12.00.$$

Hence, both statisticians A and B are correct but their ranges differ due to use of two different definitions of quartiles.

$$(b) \text{ sample mean} = \sum \frac{x_i}{15} = 213.47$$

$$\text{sample variance} = \frac{1}{14} \{ \sum x_i^2 - 15 \bar{x}^2 \} = \frac{1605.73}{14} = 114.70$$

$$(c) X \sim N(213.47, 114.70)$$

$$P(X > 230) = P\left(Z > \frac{230 - 213.47}{\sqrt{114.7}}\right) = P(Z > 1.543)$$

Using the tables and using linear interpolation between 1.54 and 1.55, we get the values as

$$P(Z > 1.543) = 1 - 0.93846 = 0.06154$$

(d) Using empirical distribution, we have 15 observations and only one observation is greater than 230. Hence probability = $\frac{1}{15} = 0.067$.

Thus, we can see that both the probabilities are close.

[9]

Q. 2) We use the formula, $\gamma_1 = \frac{\mu_3}{\sigma^3}$

Let X be the random variable representing the mass of an elephant.

We know that

$$\Pr(X = 900) = \frac{5}{12}$$

$$\Pr(X = 1,000) = \frac{4}{12} = \frac{1}{3}, \text{ and}$$

$$\Pr(X = 1,396) = \frac{3}{12} = \frac{1}{4}$$

$$E(X) = 900 \times \frac{5}{12} + 1,000 \times \frac{1}{3} + 1,396 \times \frac{1}{4} = 1,057.33$$

$$\text{Var}(X) = \mu_2 = (900 - 1,057.33)^2 \times \frac{5}{12} + (1,000 - 1,057.33)^2 \times \frac{1}{3} + (1,396 - 1,057.33)^2 \times \frac{1}{4} = 40,083.56$$

$$\sigma^3 = \text{Var}(X)^{1.5} = 80,25,081.09$$

$$\mu_3 = (900 - 1,057.33)^3 \times \frac{5}{12} + (1,000 - 1,057.33)^3 \times \frac{1}{3} + (1,396 - 1,057.33)^3 \times \frac{1}{4} = 80,25,685.58$$

$$\text{Thus, } \gamma_1 = \frac{\mu_3}{\sigma^3} = 1.00$$

[3]

Q. 3) D = Has H1N1

T = Test result is positive

Given: $P(D) = 0.001$ so $P(\text{Not } D) = 0.999$

$P(T \mid \text{Not } D) = 0.05$

$P(T \mid D) = 1.00$

We need to find $P(D \mid T)$

Now, $P(T \cap D) = P(D)P(T \mid D)$

$$= 0.001$$

$P(T) = P(T \cap D) + P(T \cap \text{Not } D)$

$$= P(T \cap D) + P(\text{Not } D)P(T \mid \text{Not } D)$$

$$= 0.001 + (0.999)(0.05)$$

$$= 0.05095$$

$$P(D \mid T) = P(T \cap D) / P(T) = 0.001 / 0.05095 = 0.019627$$

[4]

Q.4) $Y = e^{2X} + 5$ implies that

$F_Y(y)$

$$= \Pr(Y \leq y)$$

$$= \Pr(e^{2X} + 5 \leq y)$$

$$= \Pr(e^{2X} \leq y - 5)$$

$$= \Pr(2X \leq \ln(y - 5))$$

$$= \Pr(X \leq \frac{1}{2} \ln(y - 5))$$

$$= F_X[\frac{1}{2} \ln(y - 5)]$$

$$= F_X[g(y)], \quad \text{where } g(y) = \frac{1}{2} \ln(y - 5).$$

[2]

Q.5) $C_X(t) = \ln M_X(t)$

$$C'_X(t) = \frac{M'_X(t)}{M_X(t)}$$

$$C_X''(t) = \frac{M_X''(t) M_X(t) - (M_X'(t))^2}{(M_X(t))^2}$$

$$\text{Now } M_X(t) = E(e^{tX})$$

Expanding the series, we have

$$\Rightarrow M_X(t) = E\left(1 + tX + \frac{t^2}{2!} X^2 + \dots\right)$$

$$\Rightarrow M_X(t) = 1 + tE(X) + \frac{t^2}{2!} E(X^2) + \dots$$

$$\Rightarrow M_X'(t) = E(X) + tE(X^2) + \dots$$

$$\Rightarrow M_X''(t) = E(X^2) + \dots$$

Now,

$$M_X(0) = 1; \quad M_X'(0) = E(X)$$

$$M_X''(0) = E(X^2)$$

Using the above values, we get

$$C_X''(0) = \frac{M_X''(0) M_X(0) - (M_X'(0))^2}{(M_X(0))^2}$$

$$\Rightarrow C_X''(0) = \frac{E(X^2) * 1 - (E(X))^2}{1} = \text{Var}[X]$$

[4]

Q.6) We can define the probability density function (pdf) of B as a system of two equations with two unknowns:

$$f(x) = c \quad \text{for } 0 \leq x \leq 8;$$

$$= \frac{k}{7} e^{-\frac{x}{7}}, \quad \text{for } x \geq 8$$

$$\text{At } x = 8, \text{ we have, } f(8) = \frac{k}{7} e^{-\frac{x}{7}} = c$$

We also know that the area underneath a pdf must equal 1.

$$\text{Thus, } 8c + \int_8^{\infty} k e^{-\frac{x}{7}} / 7 dx = 1$$

$$\Rightarrow 8c + \left| -ke^{-x/7} \right|_8^{\infty} = 1$$

$$\Rightarrow 8c + ke^{-8/7} = 1. \text{ But } c = ke^{-8/7}/7, \text{ so } ke^{-8/7} = 7c, \text{ and therefore } 8c + 7c = 1 \text{ and so } c = 1/15.$$

Over the interval $0 \leq x \leq 8$, the area underneath the pdf is $8c = \mathbf{8/15}$

[5]

Q. 7) We first find $\mu = 0.23(2) + 0.34(5) + 0.43(7) \Rightarrow \mu = 5.17$.

$$\text{Now we find } \sigma^2 = 0.23(2-5.17)^2 + 0.34(5-5.17)^2 + 0.43(7-5.17)^2 = \sigma^2 = 3.7611.$$

$$\text{Now } S_n'^2 = \sum (X_i - \bar{X})^2 / n = (n-1)S_n^2 / n$$

$$E(S_n'^2) = E((n-1)S_n^2/n) = (n-1)E(S_n^2)/n = (n-1)\sigma^2/n$$

$$\text{So for } n = 13, E(S_{13}'^2) = 12\sigma^2/13 = 3.4718.$$

$$\text{Thus, the bias of } S_n'^2 \text{ is } E(S_n'^2) - \sigma^2 = 3.4718 - 3.7611 = \mathbf{-0.2893}$$

[3]

Q. 8)

$$(a) E(X) = \int_{\beta}^{\infty} \frac{x}{\theta} e^{\frac{-(x-\beta)}{\theta}} dx$$

let $y = x - \beta$; thus

$$E(X) = \int_0^{\infty} \frac{y+\beta}{\theta} e^{\frac{-y}{\theta}} dy$$

$$= \int_0^{\infty} \frac{y}{\theta} e^{\frac{-y}{\theta}} dy + \beta \int_0^{\infty} \frac{1}{\theta} e^{\frac{-y}{\theta}} dy$$

$$= \theta + \beta$$

$$\text{Now, } E(X^2) = \int_{\beta}^{\infty} \frac{x^2}{\theta} e^{\frac{-(x-\beta)}{\theta}} dx$$

let $y = x - \beta$; thus

$$E(X^2) = \int_0^{\infty} \frac{(y+\beta)^2}{\theta} e^{\frac{-y}{\theta}} dy$$

$$= \int_0^{\infty} \frac{y^2}{\theta} e^{\frac{-y}{\theta}} dy + \beta^2 \int_0^{\infty} \frac{1}{\theta} e^{\frac{-y}{\theta}} dy + 2\beta \int_0^{\infty} \frac{y}{\theta} e^{\frac{-y}{\theta}} dy$$

$$= 2\theta^2 + \beta^2 + 2\beta\theta = \theta^2 + (\theta + \beta)^2.$$

$$S.d. (X) = \sqrt{E(X^2) - (E(X)^2)} = \sqrt{\theta^2} = \theta$$

(b) We use two moments, since there are two parameters to estimate:

We are given that there are 50 pieces of data and that their sum is 8200. Thus,

$$E(X) = 8200/50 = 164. \text{ Likewise, } E(X^2) = \sum(x_i^2)/50 = 40000.$$

Thus, we set up the following system of equations:

$$A: 164 = \theta + \beta;$$

$$B: 40000 = \theta^2 + (\theta + \beta)^2.$$

We know that $(\theta + \beta)^2 = 164^2 = 26896$, so $\theta^2 = 40000 - 26896 = 13104$, and thus $\theta = 114.47$

Hence, $\beta = 164 - \theta = 49.52$.

[7]

- Q. 9)** Let X be the outcome from the one dice rolled. Since N is determined by the value of X ,
 $E(N) = E(X)$ and $\text{Var}(N) = \text{Var}(X)$.

$$\text{First, we find } E(X) = (1/8)(1+2+3+4+5+6+7+8) = 4.5 = E(N)$$

$$\text{Now we find } \text{Var}(X) = (1/8)((1-4.5)^2 + (2-4.5)^2 + (3-4.5)^2 + (4-4.5)^2 + (5-4.5)^2 + (6-4.5)^2 + (7-4.5)^2 + (8-4.5)^2) = 5.25 = \text{Var}(N)$$

Let S be the payoff in each individual round. Then

$$\text{Var}(S) = E(N) \cdot \text{Var}(X) + \text{Var}(N) \cdot E(X)^2 = 5.25(4.5 + 4.5^2) = 129.93.$$

[4]

- Q.10)** First, we find the mean loss per insured.

Let N = frequency of losses to individual insured; X = magnitude of losses; and S = total loss to individual insured.

$$E(S) = E(N) \cdot E(X) = 0.1 \cdot 500 = 50.$$

For n insureds, the expected loss is therefore $50n$, and 1.5 times the expected loss is $75n$.

Now we find the variance of the loss per insured.

$$\begin{aligned} \text{Var}(S) &= E(N) \cdot \text{Var}(X) + \text{Var}(N) \cdot [E(X)]^2 \\ &= 0.1 \cdot (1000)^2 + 0.1 \cdot (500)^2 \\ &= 0.1 \cdot 1,250,000 \end{aligned}$$

The variance for n insureds is thus $125000n$, and the standard deviation is $353.55\sqrt{n}$.

We want to find n such that $\Pr(\text{Total loss} \geq 75n) \leq 0.05$.

For minimum n , we need $\Pr(\text{Total loss} \geq 75n) = 0.05$

By the central limit theorem assuming total losses follow normal distribution

$$\Rightarrow \Pr(\text{Total loss} \geq 75n) = 0.05$$

$$\Rightarrow \Pr[z \geq (75n - 50n)/(353.55\sqrt{n})] = 0.05$$

$$\Rightarrow 1 - \Phi((75n - 50n)/(353.55\sqrt{n})) = 0.05$$

$$\Rightarrow \Phi(0.07071\sqrt{n}) = 0.95$$

$$\Rightarrow 0.07071\sqrt{n} = 1.645$$

$$\Rightarrow n = (1.645 / 0.07071)^2 = 541.2,$$

which we round up to **542 insureds**.

[5]

Q.11) $MSE = E[(X^2 - \theta^2)^2]$
 $= E(X^4 - 2X^2\theta^2 + \theta^4)$
 $= 24\theta^4 - 2(2\theta^2)\theta^2 + \theta^4 = 21\theta^4$ (using $E(X^r) = \Gamma(1+r) * \theta^r$ from the tables)

Any alternative solution is equally acceptable

[3]

Q.12) Next year, the severities will change to 0.6 of their previous amounts; thus, the severities will become

$$0.6*350 = 210, 0.6*460 = 276, 0.6*900 = 540, 0.6*3400 = 2040.$$

With the 200 deductible, the payout for each magnitude of loss will be 10, 76, 340, and 1840, respectively.

Thus, the expected payout per claim is $10*0.4 + 76*0.3 + 340*0.2 + 1840*0.1 = 278.8$.

Total expected payout is $E(N)*(Expected\ payout\ per\ claim) = 5667*278.8 = \mathbf{1,579,959.6}$.

[3]

Q.13) (a) If we have m values between y and z and p values above z in addition to the n known values, then the likelihood function is given by

$$L(\theta) = [P(X < y)]^n \times [P(y < X < z)]^m \times [P(X > z)]^p$$

Thus, using above, if we have 9 losses less than 10, 6 losses between 10 and 25 and 5 losses above 25, then the likelihood function is given by:

$$L(\theta) = [P(\theta < X \leq 10)]^9 * P(10 < X \leq 25)]^6 * [P(X > 25)]^5$$

$$L(\theta) = (1 - \theta/10)^9 * \{(1-\theta/25) - (1-\theta/10)\}^6 * \{1-(1-\theta/25)\}^5$$

Which is proportional to $(10 - \theta)^9 * \theta^{11}$

$$\text{Log}\{L(\theta)\} = 9\text{Log}(10 - \theta) + 11 \text{Log}(\theta)$$

Differentiating w.r.t θ

$$\frac{\partial}{\partial \theta} \log L(\theta) = -9/(10 - \theta) + 11/\theta$$

Equating this to zero, we get

$$110 = 20\theta$$

$$\hat{\theta} = 5.5$$

Differentiating $\frac{\partial}{\partial \theta} \log L(\theta)$ w.r.t θ again, we get

$$\frac{\partial^2}{\partial \theta^2} \log L(\theta) = -9/(10 - \theta)^2 - 11/\theta^2$$

We can see that $\frac{\partial^2}{\partial \theta^2} \log L(\theta)$ is always negative, which implies $\hat{\theta} = 5.5$ is the MLE of θ

(b) Substituting $\hat{\theta} = 5.5$, we have $\frac{\partial^2}{\partial \theta^2} \log L(\hat{\theta}) = -0.80808$

$$CRLB = \frac{1}{E\left\{\left[\frac{\partial}{\partial \theta} \log L(\theta)\right]^2\right\}} \text{ or } \frac{1}{-E\left[\frac{\partial^2}{\partial \theta^2} \log L(\theta)\right]}$$

CRLB = 1.2375, as the $\frac{\partial}{\partial \theta} \log L(\theta) = 0$ at $\hat{\theta} = 5.5$

(c) $\hat{\theta} \approx N(\theta, CRLB)$

95% CI is $\hat{\theta} \pm 1.96\sqrt{CRLB} = (3.32, 7.68)$

[10]

Q.14) For the Poisson distribution, the mean, λ , is estimated as $230/1000 = 0.23$.

Number of days work missed	Poisson Probability	Expected Number of Workers	Observed Number of Workers	$(O-E)^2 / E$
0	0.794534	794.53	818	0.69
1	0.182743	182.74	153	4.84
2	0.021015	21.02	25	0.76
3 or more	0.001708	1.71	4	3.08
Total	1	1000	1000	$\chi^2 = 9.37$
Marks	1.5 marks			2 marks

The χ^2 distribution has 2 degrees of freedom because there are four categories and the Poisson parameter is estimated (d.f. = 4 – 1 – 1 = 2)

The critical value for a chi-square test with two degrees of freedom and 5% level of significance is 5.99. Since Chi Square calculated is greater than Chi Square tabulated, null hypothesis is rejected.

[5]

Q.15) (i) Missing entries are:

d.f. between regions = $3-1 = 2$

d.f. total = $30-1 = 29$

d.f. Residual = 27 by subtraction

SS = 5992.8 by subtraction

MSS = 2176.4 being $4352.8/2$.

MSS = 221.96 being $5992.8/27$.

Observed $F = 9.8056$ on 2 and 27 d.f.

From tables the 5% critical point is 3.354

Significant at 5%, so there is quite strong evidence of a difference in the mean claim amounts for the three regions (significant at even 2.5% or 1% critical point, so answer do not change)

(ii) 95% confidence interval for $(\mu_A - \mu_B)$ is

$$(\bar{y}_A - \bar{y}_B) \pm t_{\frac{5\%}{2}, 27} \hat{\sigma} \sqrt{\frac{1}{10} + \frac{1}{10}}$$

giving

$$(155.48 - 164.68) \pm 2.052 \sqrt{221.96} \sqrt{\frac{1}{10} + \frac{1}{10}} \text{ as } \hat{\sigma} = \text{MSS}$$

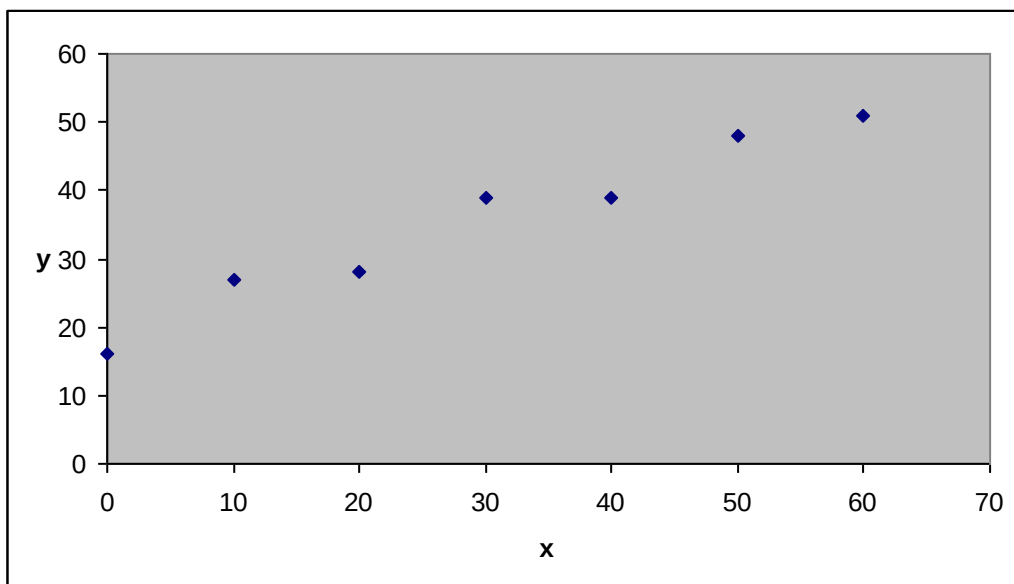
$$-9.2 \pm 13.67 = (-22.87, 4.47)$$

The interval contains zero indicating no difference between the underlying means for regions A and B.

The significant result of the F -test clearly comes from region C mean being much lower than the region A and B means

[7]

Q.16) (a)



(b) (i)

								Total
x	0	10	20	30	40	50	60	210
y	16	27	28	39	39	48	51	248
xy	0	270	560	1,170	1,560	2,400	3,060	9,020
x ²	0	100	400	900	1,600	2,500	3,600	9,100

$$S_{xy} = \sum x_i y_i - \sum x_i \sum y_i / n = 9020 - 210 * 248 / 7 = 1,580$$

$$S_{xx} = \sum x_i^2 - (\sum x_i)^2 / n = 9100 - 210^2 / 7 = 2,800$$

$$\hat{b} = S_{xy} / S_{xx} = 1580 / 2800 = 0.564$$

$$\hat{a} = \bar{y} - \hat{b} \bar{x} = (248 / 7) - 0.564 * (210 / 7) = 18.5$$

$$y = 18.5 + 0.564x$$

b(ii) Draw a line joining the two (x, y) co-ordinates: (0, 18.5) and (60, 52.4)

{OR any other co-ordinate provided the value of y is correctly calculated and plotted, against the value of x}

$$(c) (i) \sigma^2 = (S_{yy} - S_{xy}^2 / S_{xx}) / (n - 2)$$

$$S_{yy} = \sum y_i^2 - n \bar{y}^2 = 9716 - 248^2 / 7 = 929.71$$

$$\text{Thus, } \sigma^2 = (929.71 - 1580^2 / 2800) / 5 = 7.63$$

$$se(\hat{b}) = (\sigma^2 / S_{xx})^{1/2} = (7.63 / 2800)^{1/2} = 0.0522$$

$$95\% \text{ confidence interval for } b \text{ is } \hat{b} \pm \{t_{0.025, 5} * se(\hat{b})\}$$

$$= 0.564 \pm (2.571 * 0.0522)$$

$$\text{So } 95\% \text{ confidence interval is } (0.430, 0.698)$$

c(ii) The 95% two-sided confidence interval does not contain the value "1", so the two-sided test conducted at 5% level results in H_0 being rejected.

(d) (i) Difference between 8:40 am and 7 am is 100 minutes.

$$\hat{z} = \text{Time available} - \text{Time taken}$$

$$= (100 - x) - y$$

$$= (100 - x) - (\hat{a} + \hat{b} x)$$

$$= 81.5 - 1.564x$$

$$(ii) \hat{z} = 81.5 - 1.564x$$

For $\hat{z} = 0$, $x = 81.5 / 1.564 = 52.11$, taken equal to 52 minutes as approximation.

The latest time is therefore 7:52 am.

[14]

- Q.17)** H_0 : there is no difference in the I.Q before and after the training programme
 H_1 : there is difference in the I.Q before and after the training programme

Candidate	I.Q before training (x)	I.Q after training (y)	d = (y-x)	d - $\bar{d}$	(d - $\bar{d}$) ²
1	110	115	5	2.5	6.25
2	105	117	12	9.5	90.25
3	111	109	-2	-4.5	20.25
4	125	124	-1	-3.5	12.25
5	120	122	2	-0.5	0.25
6	115	114	-1	-3.5	12.25
Total			$\Sigma d = 15$	0	$\Sigma(d - \bar{d})^2 = 141.5$

$$\bar{d} = 15/6 = 2.5$$

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = \sqrt{\frac{141.5}{5}} = 5.32$$

$$\text{under } H_0, \frac{\bar{d}}{s/\sqrt{n}} \sim t_{n-1}$$

$$\text{thus, } t_{\text{calculated}} = 1.15$$

$$t_{\text{tabulated}} = t_{0.025; 5} = 2.571$$

Since $t_{\text{calculated}} < t_{\text{tabulated}}$, H_0 is accepted. Hence, there is no difference in the I.Q before and after the training programme

(b) We know that $\frac{1}{2} \log_e \left(\frac{1+r}{1-r} \right)$ is approximately normal distributed with mean $\frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right)$

and standard deviation $\frac{1}{\sqrt{n-3}}$

where r is the sample correlation coefficient and ρ is the population correlation coefficient

$$95\% \text{ C.I. for } \frac{1}{2} \log_e \left(\frac{1+r}{1-r} \right) \text{ is given by } \frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right) \pm 1.96 \frac{1}{\sqrt{n-3}}$$

$$\Rightarrow \frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right) \text{ lies between } \frac{1}{2} \log_e \left(\frac{1+r}{1-r} \right) \pm 1.96 \frac{1}{\sqrt{n-3}}$$

$$\Rightarrow \frac{1}{2} \log_e \left(\frac{1+\rho}{1-\rho} \right) \text{ lies between } \frac{1}{2} \log_e \left(\frac{1.7}{0.3} \right) \pm 1.96 * (1/7) = 0.5873, 1.1473$$

$$\Rightarrow \rho \text{ lies between } 0.528 \text{ and } 0.817$$

$$(c) f(x, y) = \frac{1}{27}(2X + Y) \quad X = 0, 1, 2$$

$$Y = 0, 1, 2$$

Joint probability distribution $f(x, y)$ of X and Y is given by table

X/Y	0	1	2
0	0	1/27	2/27
1	2/27	3/27	4/27
2	4/27	5/27	6/27
$f_Y(Y)$	6/27	9/27	12/27

The marginal probability distribution of Y is given by $f_Y(y) = \sum_x f(x, y)$

The conditional distribution of X given $Y=y$ is given by the table below

$$f_{X/Y}(X = x / Y = y) = \frac{f_{xy}(x, y)}{f_Y(y)}$$

X=x/Y=y	0	1	2
0	0	1/9	2/12
1	2/6	3/9	4/12
2	4/6	5/9	6/12

[12]

Total Marks 100
